IMMERSIONS INDUCE HOMEOMORPHISMS ONTO LOCALLY CLOSED SUBSETS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original lemma 03H4: sources/stacks-chow/tags.txt, source lines 4553–4553. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02YT: sources/stacks-chow/tags.txt, source lines 3888–3888. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02YB: sources/stacks-chow/tags.txt, source lines 3870–3870. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $Z \to X$ be an immersion of algebraic spaces. Then $|Z| \to |X|$ is a homeomorphism of $|Z|$ onto a locally closed subset of $|X|$.*

Proof. Let U be a scheme and $U \to X$ a surjective étale morphism. Then $Z \times_X U \to U$ is an immersion of schemes, hence gives a homeomorphism of $|Z \times_X U|$ with a locally closed subset T' of $|U|$. By Lemma 03H4 the subset T' is the inverse image of the image T of $|Z| \to |X|$. The map $|Z| \to |X|$ is injective because the transformation of functors $Z \to X$ is injective, see Spaces, Section 02YT. By Topology, Lemma 02YB we see that T is locally closed in $|X|$. Moreover, the continuous map $|Z| \to T$ is a homeomorphism as the map $|Z \times_X U| \to T'$ is a homeomorphism and $|Z \times_Y U| \to |Z|$ is submersive. □

Date: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.